Tech Interview Prep:

# SQL vs. NoSQL and Read Replica Scaling

Section 1: SQL vs. NoSQL: Beyond the Buzzwords

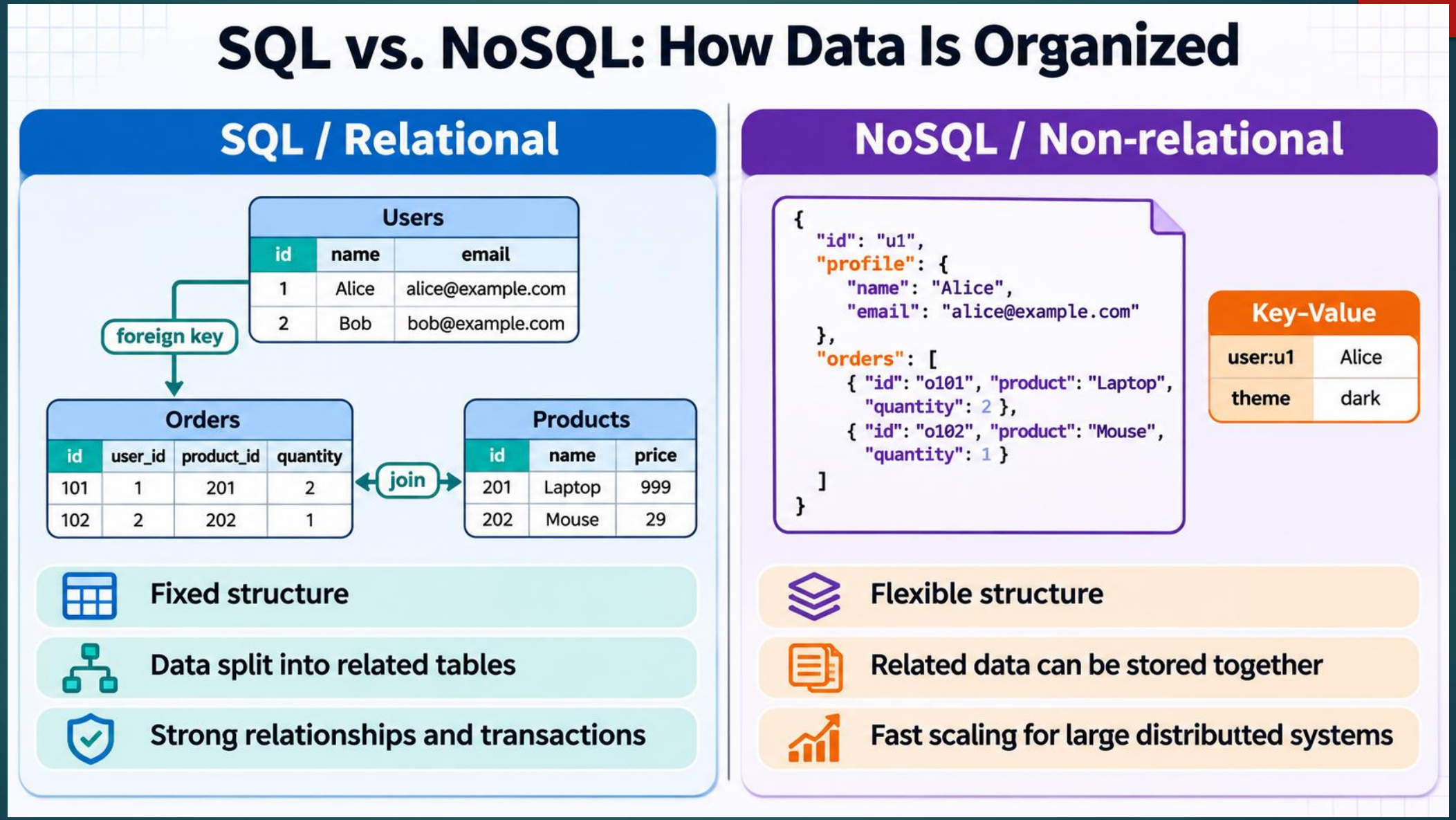
Section 2: Read Replicas & Scaling the Read Path

Section 3: Synthesis & Architectural Rule of Thumb

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CLASS: AD371 (APPLICATION ARCHITECTURE)

# Section 1.1: SQL vs. NoSQL Comparison



# Consistency and Scaling Trade-offs

## ACID



- Strong consistency
- Transactions stay correct
- Best for payments and orders

## BASE



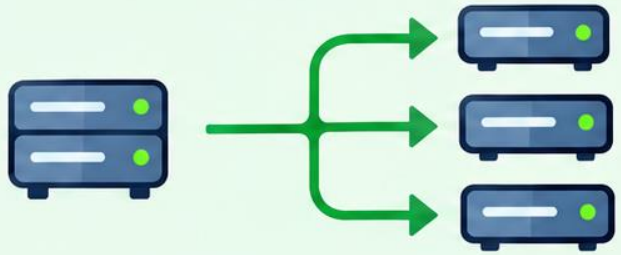
- Eventual consistency
- Copies may briefly differ
- Best for large distributed systems

## Vertical scaling (scale up)



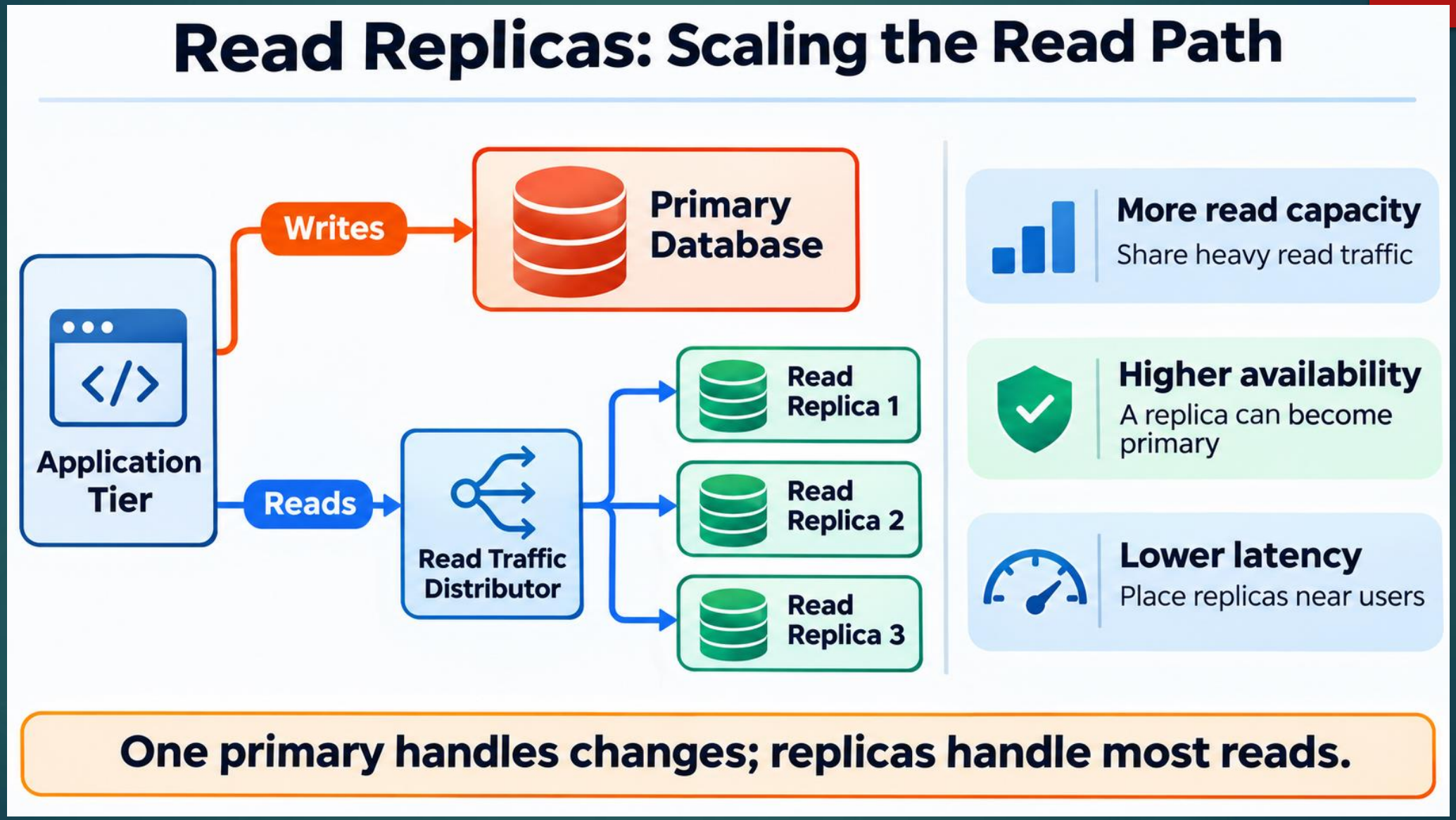
Add more CPU or memory to one machine

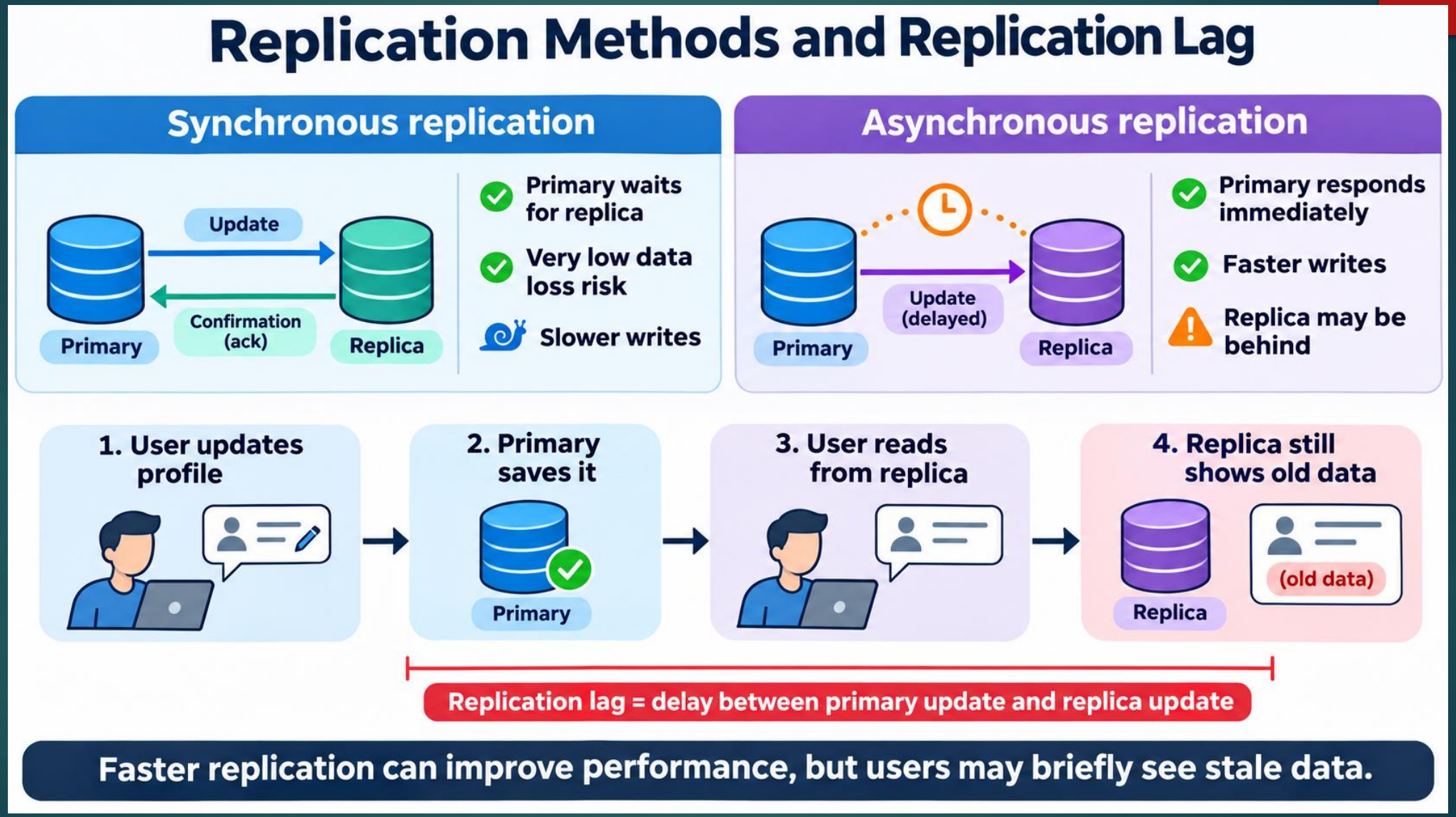
## Horizontal scaling (scale out)

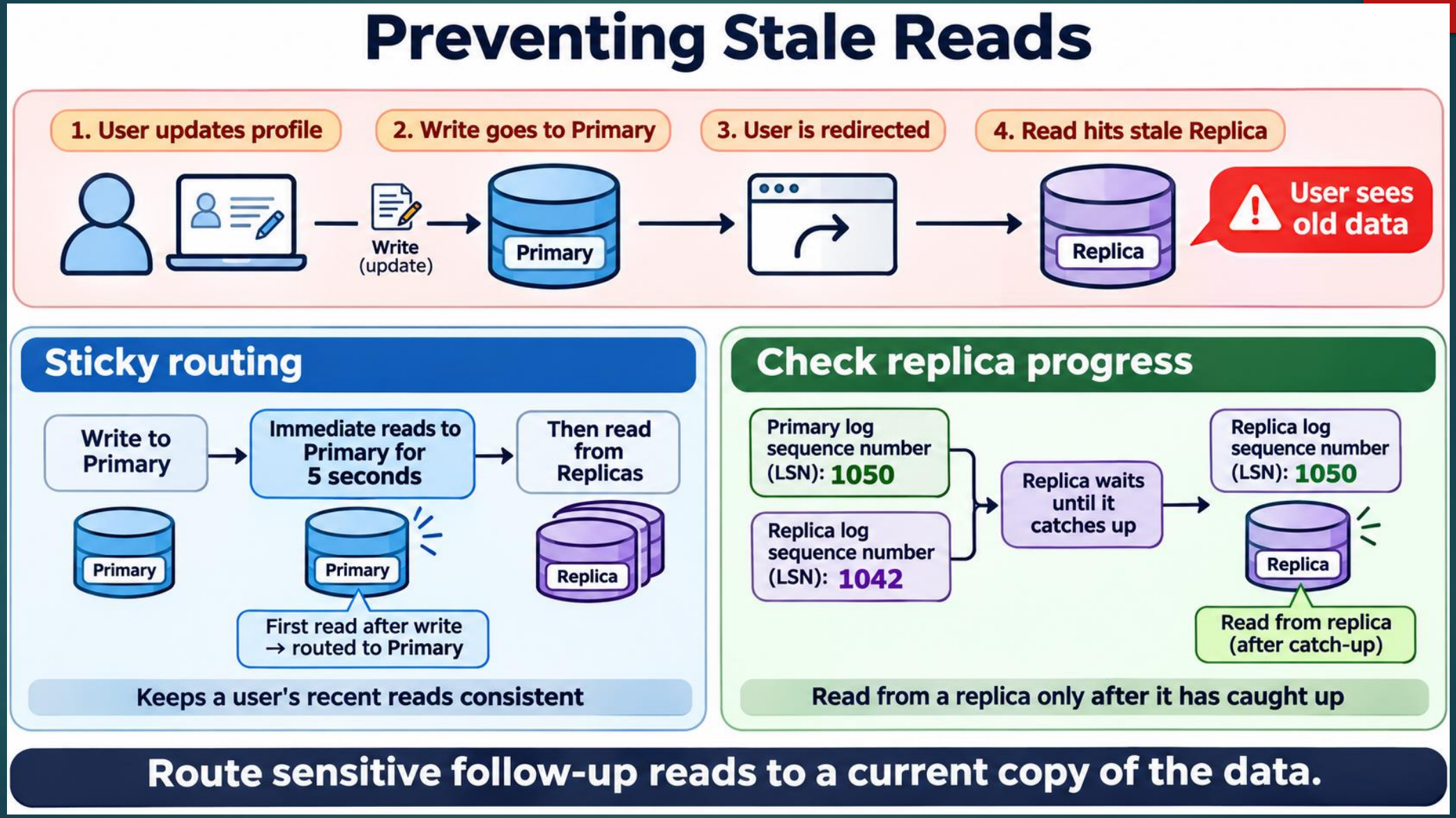


Add more machines and distribute the data

**Choose the trade-off that matches the application.**


Read Replicas: Scaling the Read Path
Writes
Primary Database
Application Tier
Reads
Read Traffic Distributor
Read Replica 1
Read Replica 2
Read Replica 3
More read capacity
Share heavy read traffic
Higher availability
A replica can become primary
Lower latency
Place replicas near users
One primary handles changes; replicas handle most reads.


Replication Methods and Replication Lag
Synchronous replication
Update
Confirmation (ack)
Primary
Replica
Primary waits for replica
Very low data loss risk
Slower writes
Asynchronous replication
Update (delayed)
Primary
Replica
Primary responds immediately
Faster writes
Replica may be behind
1. User updates profile
2. Primary saves it
Primary
3. User reads from replica
4. Replica still shows old data
Replica
(old data)
Replication lag = delay between primary update and replica update
Faster replication can improve performance, but users may briefly see stale data.


Preventing Stale Reads
1. User updates profile
2. Write goes to Primary
3. User is redirected
4. Read hits stale Replica
Write (update)
Primary
Replica
User sees old data
Sticky routing
Write to Primary
Immediate reads to Primary for 5 seconds
Then read from Replicas
Primary
Primary
Replica
First read after write → routed to Primary
Keeps a user's recent reads consistent
Check replica progress
Primary log sequence number (LSN): 1050
Replica log sequence number (LSN): 1042
Replica waits until it catches up
Replica log sequence number (LSN): 1050
Replica
Read from replica (after catch-up)
Read from a replica only after it has caught up
Route sensitive follow-up reads to a current copy of the data.

# Section 3: Synthesis & Architectural Rule of Thumb

- Strong Consistency vs. Eventual Consistency
- Primary Database  vs. Replicas
- Preventing Stale Reads (Sticky Routing / Check Replica Progress)